

The Ohio State University Imaging Sciences Laboratory (ISL)

Bruce Atwood, P. L. Byard, Darren L. DePoy, Jerry A. Mason, Paul Martini, Thomas P. O'Brien, Richard W. Pogge, Daniel Pappalardo, and Andrew Stephens

Imaging Sciences Laboratory, Department of Astronomy, Ohio State University, 174 W. 18th Ave., Columbus, Ohio 43210-1106.

Abstract

The ISL is a successful astronomical instrumentation program that has completed three major instruments (an imaging Fabry-Perot spectrometer and two infrared imager/spectrometers) and many smaller projects since 1987. We have developed the capabilities to perform all aspects of instrument design and construction and a range of unique skills and methods. We maintain a permanent staff that currently consists of two scientists specializing in optical design and detector systems, a senior mechanical engineer, a programmer, an electronic engineer, a mechanical designer, two machinists, and a lab assistant. Instrumentation projects also draw upon faculty and graduate student effort.

We will describe the ISL staff and philosophy and give brief descriptions of recent projects, detector electronics, optical design for two-color cameras, cooling of InSb arrays, and our instrument control system.

1. INTRODUCTION

Ohio State has had an active astronomical instrumentation program for more than twenty years. Early efforts were organized, as they are in many university environments, for the construction of one specific instrument. First-time costs were a major factor in design choices and each new project was begun largely without regard for the details of previous efforts. Most notable of these early efforts are the IDS photon counting¹ system and the CCDS spectrograph both of which have had substantial success on the 1.8-m Perkins telescope at Lowell Observatory. About ten years ago the department, recognizing that their participation in the LBT project (then called the Columbus Project) as well as other instrumentation projects would benefit from a more systematic approach, formed what is now called the Imaging Sciences Laboratory (ISL). The ISL, with a significant number of permanent employees, has implemented a complete and stable instrument development environment where previously tested designs serve for a large fraction of each new project. Care was taken to define an overall system architecture that would be long-lived and extensible. Problems such as electronics packaging and cabling were addressed and internal standards were developed and are enforced to allow quick development of new projects. Robust sub-systems were developed for detector control, data handling, instrument control, telescope control, engineering data logging, and user interface. Equally important with these hardware designs and standards was the development of an infrastructure and culture to allow the ISL to tackle the design and fabrication of new system with confidence. We viewed it as essential that we have access to first rate software for optical design (CODE-V), mechanical analysis (ALGOR) and digital design (ALTERA). All fabrication, mechanical, electronic, optical, is done from full documentation, thus preserving designs for future use and allowing analysis before construction.

Modularity is always a goal. Since a module is, almost by definition, a unit with well defined and relatively simple connections with other modules, modular design facilitates reuse and analysis when compared to strategies that tightly couple many elements of a single subsystem. Modularity does not necessarily lead to designs with the lowest first-time cost. More connectors, for example, are required and since interfaces must be clean and simple, designs where a single element serves several different functions, are discouraged.

We have developed a number of guidelines, some mundane some not, that we feel make for better instruments. Two examples, illustrating the range of subjects covered, are how modules are interconnected and how changes in temperature are to be accommodated. All assemblies must be easy to remove and replace. Modules are joined with: screws, not rivets; connectors not solder joints, command lines not multi-dimensional data bases. As a second example

of our guidelines, we insist that, wherever possible and especially in cryogenic instruments, the strain from differential thermal expansion is located in elastic flexure. To do otherwise means that a module may behave differently each time it is cooled, making it difficult or impossible to predict how it will work.

Finally, we *actively* seek to *minimize* the amount and importance of software in each project. Within an instrument, software is viewed as a tool for using hardware. Since software is the part of a project which most frequently fails we minimize risk by minimizing the amount of software.

We take the construction of 5 major instruments with successful use on 9 different telescopes at 5 different observatories to be strong evidence of the success of our approach

2. Facilities

The Department of Astronomy, including the ISL, is currently located in Smith Lab on the Columbus campus of The Ohio State University. The 900-square foot basement machine shop is equipped with three lathes, two mills, band saws and a welding facility which routinely produces vacuum/cryo welds in aluminum up to 19 mm thick and stainless steel as thin as 60 microns. Fifth floor labs include a 600-square foot instrument assembly and test area which also houses our helium leak detector. The 600-square foot electronics lab includes three work areas for electronics assembly and testing and an area for electronic stock. A third 600-square foot lab house our fume hood, 63-centimeter-diameter stainless steel high-vacuum evaporation system, and a 4 by 8 foot optical table. A 300-square foot lab houses two copies of our ICIMACS² instrument control system while a second 300-square foot lab is used for detector system development and houses our clean-bench.. Our electronic engineer, mechanical designer, and lab assistant share a 550-square foot office with our drawing files and E-size plotter. The ISL also occupies four staff offices.

The ISL's fully networked computing facilities include numerous PC and three Sun SPARCstations. Supported software includes Code V for optical design, Auto-Cad for mechanical drawing, the Algor finite element analysis package, the Altera MAX+plus II Erasable Programmable Logic Device development package, the ORCAD schematic capture and PADS printed circuit design system, Matlab, Mathcad, Lablog and numerous compilers and utilities.

Major laboratory instruments include a helium mass spectrometer leak detector, digital and analog oscilloscopes, 100kHz spectrum analyzer, two twenty-four-channel analog data acquisition systems, monochromator, a spectrophotometer for absolute reflectivity and transmission measurements between 300 and 1000nm, a four by eight foot optical table and a full complement of laboratory test equipment.

2. STAFF

The director of the ISL, has more than 30 years experience with electronics and optical instruments including more than 11 years at Cerro Tololo Inter-American Observatory where his title was Senior Support Scientist. A second research scientist has been working on astronomical instrumentation for more than 35 years, with the last eight devoted almost exclusively to optical design. Our chief mechanical engineer has over 14 years experience in the design and analysis of mechanisms and structures, including both high temperature and cryogenic applications. In addition to these senior staff members, the ISL has seven other permanent salaried employees: a machinist with 12 years experience in scientific shops who has extensive training in metal joining technology, a machinist with 15 years experience in a combination of scientific and industrial shops, a mechanical designer with extensive experience with both structures and mechanisms, an design engineer, a computer specialist, a lab assistant and researcher, and a half-time electronics technician. Approved additional positions include a staff member for either electronic or mechanical fabrication and a senior electronic engineer to be hired next fiscal year.

3. CAPABILITIES

The specific combination of ISL facilities and staff allows us to produce fully engineered and characterized mechanical, optical, electronic, and detector systems for ground-based optical and IR astronomy. The staff is large enough to provide excellence in all required areas of design. Since the staff is permanent we can view each new instrument not only as an

isolated project but also as an opportunity to make a long-term contribution to our infrastructure. New instrument always draw very heavily on previous ones.

4. INSTRUMENTS

Imaging Fabry-Perot Spectrometer (IFPS)

The OSU IFPS is an optical CCD imaging system optimized for high-throughput operation in the 320 to 1000 nm wavelength range. It has found use as both an imaging Fabry- Perot spectrograph and as a conventional 5:1 focal-reducing imaging camera. Mounted at the $f/17.5$ Cassegrain focus of the 1.8 m-Perkins telescope, it provides an unvignetted square field of view of 6.5 arcminutes with an image scale of 0.5 arcseconds/pixel. All glass/air surfaces have been custom AR coated to provide the high throughput across the entire wavelength range of operation. Instrument functions are fully automated under the ICIMACS system. The internal instrument layout follows an optical bench design, allowing relatively rapid modification of the optical configuration to accommodate different input beams (e.g. , for $f/15$ versus $f/17.5$), as well as providing the capability to add additional equipment to the optical train for special experiments. Instrument throughput, including telescope and detector, is about 25% in Fabry-Perot mode, increasing to about 45% in direct-imaging mode.

A set of narrow-band order-separation filters for most Galactic nebular lines, and redshifted $H\alpha$ +[N II], [S II], and [O III] filter sets are available. User filters may be mounted using cartridges to accommodate both 64-mm and 50-mm square filters, or custom holders can be made for filters up to 64 mm across. A parfocal UBVRI filter set is also available for broad- band direct imaging, and the order-separation filters may be used as stand-alone narrow-band filters for direct imaging.

A unique feature of the IFPS is its near-UV capability, especially for Fabry-Perot and narrow-band imaging projects. Our "UV" etalon has sufficient resolving power to split the [O II] 372.7 nm, 372.9 nm doublet lines, an important diagnostic of nebular electron density. It can also obtain emission-line images in the high-ionization [Ne V] 342.6 nm line, a key tracer of hard-photon ionization in nebulae. Near-UV filters must be red-blocked for use with CCD detectors, and traditional self-blocked UV filters are limited to a peak throughput to less than 40%. We use unblocked filters in conjunction with a crystal CuSO_4 filter that can be flipped into the collimated beam to achieve narrow-band transmission of >70%, which combined with a custom UV-to-IR lens design and special anti-reflection coatings gives the IFPS high near-UV efficiency for F-P and narrow-band imaging applications. These features also make it a superb U- band direct imaging system.

Ohio State Infrared Imager/Spectrometer

OSIRIS is a general purpose near infrared (0.9 to 2.5 microns) instrument with both imaging and spectroscopic capabilities. In imaging mode infrared radiation is re-imaged onto the detector array by a fixed collimator and one of up to four selectable camera lens assemblies. We currently have an $f/3$ set that gives 1.5" pixel on a 1.8-m telescope and an $f/7$ camera that gives 0.6"/pixel. There is also a silicon lens set that produces images of the pupil on the detector making alignment of the instrument with the optical axis of a telescope straightforward. Two 10- position filter wheels allow for 20 individual round 30-mm filters, grisms, cross-dispersers, or dark stops.

In spectroscopic mode, a grating is moved into the collimated optical beam, replacing a folding flat mirror, and a slit is introduced into the telescope focal plane, replacing the field stops used in imaging. The resolution depends on the choice of camera lens and the slit width. With the $f/3$ cameras and a 4.5" wide slit OSIRIS has a resolution of 570, which conveniently disperses an entire atmospheric window across the detector array. The $f/7$ camera and a 2" slit gives a resolution of 1350. The wavelengths observed are set by the remotely-controlled grating angle and the order-sorting filter. The slit covers the entire height of the array: 6.5" at $R=570$ and 2.5" at $R=1350$. By restricting the slit height to 1' and using a grism in place of the order sorting filter we cross-disperse the beam and simultaneously observe the J, H and K atmospheric windows at $R=570$.

Although designed for the $f/17$, 1.8-m Perkins telescope, OSIRIS can be used on nearly any telescope; only minor modifications are needed, such as slight repositioning of the collimator optical elements or using different camera lenses. We can use OSIRIS on any telescope which has a beam slower than $f/17$ by simply changing the cold stop. We have

demonstrated this flexibility by using OSIRIS on the CTIO 4-m telescope (which has an $f/30$ secondary). OSIRIS was offered as a facility instrument on the CTIO 4-m and 1.5-m telescope for 18 months.

All of the control motions associated with OSIRIS are performed by stepper motors mounted in the vacuum vessel and operated at cryogenic temperatures. There are no mechanical feed-throughs, only electrical connections are penetrate the vacuum shell. This makes reconfiguration of the instrument quick and straightforward. For example, in imaging mode the focal-plane can be changed in less that 10 seconds. The change from imaging to spectroscopic modes of operation requires less that one minute.

In imaging mode the measured performance of OSIRIS on the CTIO 4-m telescope is $S/N=10$ in ten minutes for $K=19$ magnitude point sources in a 1" aperture. The performance in spectroscopic mode depends sensitively on the seeing (since the slit is 1.2" or 0.5" wide), but under typical conditions at CTIO we have measured $S/N=10$ per pixel for $K=15$ magnitude point sources in 30 minutes in the $R=570$ mode.

An agreement has been finalized for the ISL to install a CTIO supplied 1024 x 1024 pixel HgCdTe NICMOS IV array in OSIRIS and return it to Chile for at least 3 years as a facility instrument on the CTIO 4-m and 1.5-m telescopes. The return of OSIRIS to CTIO is scheduled for early 1999.

MDM-Ohio State Array Infrared Camera (MOSAIC³)

The MOSAIC is a general-purpose 1-2.5 micron imager and moderate-resolution spectrometer built as collaboration between Ohio State and the MDM Observatory. The instrument was designed to fully illuminate a 1024x1024 InSb array, but currently has a 512x1024 InSb array (supplied by NOAO and the USNO). The instrument can be used at any telescope with a beam slower than $f/7.5$ and includes cameras for two plate scales (approximately $f/7.5$ and $f/16.5$) and pupil viewing. At the MDM 2.4-m telescope the scales are 0.3" and 0.14" per pixel. Grisms provide spectroscopic dispersion with resolutions of ~ 600 to ~ 1200 (assuming a two-pixel slit) depending on the camera and grism used. All configurations of the instrument are under user control and can be selected rapidly, allowing many different observational programs to be accommodated during a single night.

MOSAIC was designed to work at the $f/7.5$ cassegrain focus of the MDM 2.4-m Hiltner telescope, but has also been used at the MDM 1.3-m and KPNO 2.1-m and 4-m telescopes. In exchange for the large detector array, the instrument is available for community use at KPNO. The optical design has a relatively small number of surfaces to optimize throughput. The terrible weather conditions during late-1997 and early- 1998 has delayed accurate measurement of the actual throughput, but an estimate is that imaging through JHK broad band filters should detect about 25% of incoming photons (including the atmosphere and telescope primary and secondary).

CCD systems

2048 by 2048 pixel CCD systems have been built for Lowell Observatory, Michigan state University, and the Wise Observatory. A 1200 by 800 pixel CCD system has been installed on the folded Schmidt camera for the Ohio State CCD spectrograph used on the Perkins telescope. In each case a version of ICIMACS controls the detector and instrument.

Andicam/Dandicam

Two nearly identical cameras for simultaneous filter imaging in the optical (320-1050 nm) and the near IR (750-2500nm) bands are nearing completion for use on the CTIO 1-m (Andicam) and South African Observatory 40-inch telescopes (Dandicam). Each of these cameras will be operated by a version of ICIMACS that will allow completely independent simultaneous optical and IR observations.

LBT

The ISL is just beginning design work on DUOS, the LBT intermediate resolution faint object optical spectrograph. The first copy of this very large instrument will be delivered in 2001. The ISL will begin the integration of the aluminizing system for the LBT primaries in mid 1999. This system will incorporate novel vacuum pumping systems, crucible type

sources, high H₂ background pressure operation, and high frequency power supplies.

5. FACILITIES

We are now in the construction phase of new space for the ISL. The 6400-square foot facility, to be located in McPherson Laboratory, will meet all the physical requirements of the ISL, and is specifically designed for the construction and testing of instruments for 8-12-meter class telescopes. In addition, the new facilities will be contiguous with the Department of Astronomy, and the very close cooperation we enjoy between observers and instrument builders will continue. Construction is scheduled to be complete in April of 1999.

References

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